

DLOPs Assignment-1 Report

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Q1(a): Deep Learning Experiments on MNIST and FashionMNIST

Key findings:

- **Model Depth vs Performance** - Increasing model depth significantly increased computational cost (FLOPs and training time), while accuracy gains were marginal or even negative for a simple dataset when trained for a small number of epochs. Indicating a large number of parameters needs time for convergence and perform poorer on a limited budget.
- **Adam vs SGD (Convergence)** - with small epochs SGD with smaller Lr gave very poor accuracy evident for both resnet-18 and resnet-50 as the model was never able to converge while Adam for same batch size, Lr, epochs and the same model gave 85+% accuracy for FashionMNIST and 90+% accuracy for MNIST getting that extra push from the momentum and adaptive learning rate.
- **Adam vs SGD (Training time)** - despite have the same FLOPs SGD showed a lower training time in comparison to Adam as Adam maintains additional moment estimates for each parameter.
- **Learning rate = 0.001 vs Learning rate = 0.0001** - A learning rate of 0.001 consistently outperformed 0.0001, as the smaller learning rate resulted in slower descent and insufficient progress within the fixed number of epochs.
- **Batch size = 16 vs Batch Size = 32** - Increasing batch size reduced training time due to improved GPU utilization for e.g. Resnet18 GPU Lr = 0.001 with batch size = 16 took 205893 ms while the same Resnet18 GPU Lr = 0.001 with batch size = 32 took 135205 ms.

Q1(b): SVM Experiments

Support Vector Machines with RBF and Polynomial kernels were trained on MNIST and FashionMNIST. Results showed that RBF kernels achieved higher accuracy but required longer training times due to increased computational complexity.

Q2: CPU vs GPU Performance on FashionMNIST

Key findings:

- **GPU vs CPU** - GPU execution reduced training time 16x times compared to CPU while achieving similar accuracy, demonstrating the importance of hardware acceleration for deep learning workloads.

- ***FLOPs vs Clock time*** - Although FLOPs increased significantly with model depth, the training time was not strictly following the FLOPs higher the model depth higher was the training time.
- ***FLOPs vs Accuracy*** - Increasing model depth increased FLOPs and training time but did not yield proportional accuracy improvements.

Assumptions: I have kept the epochs (3) same for fairness and to simulate a limited resource scenarios and increased them in case accuracy drops below 80%.
